

LABORATORY MANUAL

For

DATABASE MANAGEMENT SYSTEMS

Course Code: U25DSPC307

S.Y. BTECH

SEMESTER-III

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL NO. 2

DDL AND CONSTRAINTS IMPLEMENTATION

1. Aim

To create tables using DDL commands in MySQL and implement integrity constraints - PRIMARY KEY, FOREIGN KEY, CHECK, UNIQUE, NOT NULL and DEFAULT - and to verify that the constraints reject invalid data.

2. Problem Statement

Using the relational schema of the College Academic Management System obtained in Practical 1, create the tables *department*, *instructor*, *student*, *student_phone*, *course* and *enrollment* in MySQL with the following business rules:

- Every table must have a primary key; enrollment and student_phone use composite keys.
- Department code, course title, roll number and e-mail addresses must be unique.
- Names, designation, salary and date of birth are mandatory.
- Budget must be positive, salary between 30,000 and 3,00,000, credits between 1 and 5, marks between 0 and 100, semester between 1 and 8, gender M/F/O, date of birth not before 1995 and phone numbers must be valid 10-digit Indian mobile numbers.
- Deleting a student must delete his/her phone numbers and enrollments; deleting an instructor must leave courses unassigned; changing a department id must be propagated.

Verify each constraint by inserting valid and invalid data, and modify constraints using ALTER TABLE.

Note: The course document lists only the title of this practical. The case study, sample data and individual tasks below were prepared for this report to fulfil that title; they are illustrative and are not official university questions. All names, e-mail addresses and phone numbers are fictitious.

3. Objectives

1. To use DDL commands CREATE, ALTER and DROP for defining tables.
2. To implement entity integrity using PRIMARY KEY and NOT NULL.
3. To implement domain integrity using CHECK, UNIQUE and DEFAULT.
4. To implement referential integrity using FOREIGN KEY with referential actions.
5. To observe how MySQL reports constraint violations.

4. Software / Tools Required

Sr. No.	Software / Tool	Purpose
1	MySQL Community Server 8.0 (version 8.0.46 used)	Relational DBMS on which all SQL statements were executed
2	MySQL command-line client (mysql)	Running the SQL scripts and recording the output
3	Operating system: Ubuntu 24.04 LTS (Windows 10/11 also suitable)	Platform for the DBMS server and client
4	Text editor (VS Code / Notepad++)	Writing and saving the .sql script files

5. Theory

5.1 Data Definition Language (DDL)

DDL statements define and modify the structure (schema) of database objects. In MySQL every DDL statement causes an implicit COMMIT. The main DDL commands are:

Command	Purpose	Example
CREATE	Create a database, table, index or view	CREATE TABLE course (...);
ALTER	Add, modify or drop columns and constraints	ALTER TABLE student ADD ...;
DROP	Remove an object together with its data	DROP TABLE course;
TRUNCATE	Delete all rows quickly, keeping the structure	TRUNCATE TABLE enrollment;
RENAME	Rename a table	RENAME TABLE a TO b;

5.2 Integrity Constraints

Constraints are rules declared on columns or tables that the DBMS enforces on every INSERT, UPDATE and DELETE, so that invalid data can never be stored. A constraint can be written at column level (next to the column) or at table level using **CONSTRAINT name type (columns)**; naming a constraint makes error messages clear and allows the constraint to be dropped later.

Constraint	Rule enforced	MySQL behaviour on violation
NOT NULL	Column must always have a value	ERROR 1048: Column cannot be null
UNIQUE	No two rows may have the same value (NULLs allowed)	ERROR 1062: Duplicate entry
PRIMARY KEY	UNIQUE + NOT NULL; one per table; identifies each row (entity integrity)	ERROR 1062: Duplicate entry ... for key PRIMARY
FOREIGN KEY	Value must exist in the referenced parent key (referential integrity)	ERROR 1452 (child insert) / 1451 (parent delete)
CHECK	Boolean condition must not be FALSE (enforced from MySQL 8.0.16)	ERROR 3819: Check constraint is violated
DEFAULT	Value used when a column is omitted in INSERT	Not an error; fills value

5.3 Referential Actions

When a referenced parent row is deleted or its key is updated, the FOREIGN KEY clause decides what happens to child rows:

- **RESTRICT / NO ACTION** (default in MySQL): the parent change is rejected while child rows exist.
- **CASCADE**: the delete or key update is propagated to the child rows.
- **SET NULL**: the foreign key column of the child rows is set to NULL (column must be nullable).
- **SET DEFAULT**: parsed by MySQL but rejected by the InnoDB engine.

5.4 MySQL-Specific Notes

- CHECK constraints are parsed and *enforced* only from MySQL 8.0.16; older versions silently ignored them. A CHECK cannot use non-deterministic functions such as CURDATE().
- InnoDB automatically creates an index on every foreign key column if none exists.
- AUTO_INCREMENT generates sequential integer keys; the column must be indexed (usually the primary key).
- Constraint information is available in `information_schema.table_constraints`, `check_constraints` and `key_column_usage`.

5.5 Syntax

```
CREATE TABLE table_name (
    column_name data_type [NOT NULL] [DEFAULT value] [AUTO_INCREMENT],
    ...
    CONSTRAINT pk_name PRIMARY KEY (col, ...),
    CONSTRAINT uq_name UNIQUE (col),
    CONSTRAINT chk_name CHECK (condition),
    CONSTRAINT fk_name FOREIGN KEY (col) REFERENCES parent (col)
        [ON DELETE {RESTRICT | CASCADE | SET NULL}]
        [ON UPDATE {RESTRICT | CASCADE | SET NULL}]
);
ALTER TABLE t ADD CONSTRAINT name ...;      ALTER TABLE t DROP CHECK name;
ALTER TABLE t DROP FOREIGN KEY name;        ALTER TABLE t DROP INDEX unique_name;
```

6. Procedure / Algorithm

1. Log in to the MySQL server: `mysql -u root -p`.
2. Create the database `college_db` and select it with `USE`.
3. Decide the data type and constraints of every column from the relational schema of Practical 1 (key columns, mandatory columns, valid ranges and unique values).
4. Create the parent tables first (department), then the tables that reference them (instructor, student, course) and finally the relationship tables (student_phone, enrollment), naming every constraint.
5. Display the constraints from `information_schema` to verify that they were created.
6. Insert valid rows and confirm that they are accepted; note the effect of `DEFAULT` and `AUTO_INCREMENT`.

7. Insert invalid rows for each constraint type and record the error raised by MySQL.
8. Update and delete parent rows to observe the CASCADE, SET NULL and RESTRICT actions.
9. Use ALTER TABLE to add, drop and modify constraints and verify the final structure.

7. SQL Queries / Program

Q1: Create the database

```
DROP DATABASE IF EXISTS college_db;
CREATE DATABASE college_db;
USE college_db;
```

Q2: DEPARTMENT table: PRIMARY KEY, NOT NULL, UNIQUE, CHECK and DEFAULT

```
CREATE TABLE department (
    dept_id      INT          NOT NULL,
    dept_code    VARCHAR(10)  NOT NULL,
    dept_name    VARCHAR(40)  NOT NULL,
    building     VARCHAR(20),
    budget       DECIMAL(12,2) DEFAULT 1000000.00,
    CONSTRAINT pk_department PRIMARY KEY (dept_id),
    CONSTRAINT uq_dept_code  UNIQUE (dept_code),
    CONSTRAINT chk_budget    CHECK (budget > 0)
);
```

Q3: INSTRUCTOR table: FOREIGN KEY with referential actions and self-reference

```
CREATE TABLE instructor (
    instr_id      INT          NOT NULL,
    instr_name    VARCHAR(40)  NOT NULL,
    designation    VARCHAR(20)  NOT NULL,
    dept_id       INT,
    salary        DECIMAL(10,2) NOT NULL,
    email         VARCHAR(40)  NOT NULL,
    mentor_id     INT,
    CONSTRAINT pk_instructor PRIMARY KEY (instr_id),
    CONSTRAINT uq_instr_email UNIQUE (email),
    CONSTRAINT chk_salary    CHECK (salary BETWEEN 30000 AND 300000),
    CONSTRAINT fk_instr_dept FOREIGN KEY (dept_id)
        REFERENCES department (dept_id) ON DELETE SET NULL ON UPDATE CASCADE,
    CONSTRAINT fk_instr_mentor FOREIGN KEY (mentor_id)
        REFERENCES instructor (instr_id) ON DELETE SET NULL
);
```

Q4: STUDENT and STUDENT_PHONE tables: AUTO_INCREMENT, CHECK with IN and REGEXP

```
CREATE TABLE student (
    student_id    INT            NOT NULL AUTO_INCREMENT,
    roll_no       VARCHAR(10)    NOT NULL,
    student_name  VARCHAR(40)    NOT NULL,
    gender        CHAR(1)        NOT NULL,
    dob           DATE           NOT NULL,
    dept_id       INT            NOT NULL,
    city          VARCHAR(30)    DEFAULT 'Pune',
    email         VARCHAR(40)    NOT NULL,
    CONSTRAINT pk_student PRIMARY KEY (student_id),
    CONSTRAINT uq_roll_no  UNIQUE (roll_no),
    CONSTRAINT uq_student_email UNIQUE (email),
    CONSTRAINT chk_gender  CHECK (gender IN ('M', 'F', 'O')),
    CONSTRAINT chk_dob     CHECK (dob >= '1995-01-01'),
    CONSTRAINT fk_student_dept FOREIGN KEY (dept_id)
        REFERENCES department (dept_id) ON UPDATE CASCADE
);

CREATE TABLE student_phone (
    student_id INT            NOT NULL,
    phone      CHAR(10)      NOT NULL,
    CONSTRAINT pk_student_phone PRIMARY KEY (student_id, phone),
    CONSTRAINT chk_phone        CHECK (phone REGEXP '^[6-9][0-9]{9}$'),
    CONSTRAINT fk_phone_student FOREIGN KEY (student_id)
        REFERENCES student (student_id) ON DELETE CASCADE
);
```

Q5: COURSE and ENROLLMENT tables: composite PRIMARY KEY and ON DELETE CASCADE

```
CREATE TABLE course (
    course_id  CHAR(5)    NOT NULL,
    title      VARCHAR(40) NOT NULL,
    dept_id    INT        NOT NULL,
    credits    TINYINT    NOT NULL,
    instr_id   INT,
    prereq_id  CHAR(5),
    CONSTRAINT pk_course PRIMARY KEY (course_id),
    CONSTRAINT uq_course_title UNIQUE (title),
    CONSTRAINT chk_credits CHECK (credits BETWEEN 1 AND 5),
    CONSTRAINT fk_course_dept FOREIGN KEY (dept_id)
        REFERENCES department (dept_id) ON UPDATE CASCADE,
    CONSTRAINT fk_course_instr FOREIGN KEY (instr_id)
        REFERENCES instructor (instr_id) ON DELETE SET NULL,
    CONSTRAINT fk_course_prereq FOREIGN KEY (prereq_id)
        REFERENCES course (course_id)
);

CREATE TABLE enrollment (
    student_id INT            NOT NULL,
    course_id  CHAR(5)        NOT NULL,
    semester   TINYINT        NOT NULL,
    marks      INT,
    grade      CHAR(2),
    CONSTRAINT pk_enrollment PRIMARY KEY (student_id, course_id),
    CONSTRAINT chk_semester CHECK (semester BETWEEN 1 AND 8),
    CONSTRAINT chk_marks    CHECK (marks BETWEEN 0 AND 100),
    CONSTRAINT fk_enr_student FOREIGN KEY (student_id)
        REFERENCES student (student_id) ON DELETE CASCADE,
    CONSTRAINT fk_enr_course FOREIGN KEY (course_id) REFERENCES course (course_id)
);
```

Q6: View all constraints from the data dictionary

```

SELECT table_name, constraint_name, constraint_type
FROM   information_schema.table_constraints
WHERE  table_schema = 'college_db'
ORDER BY FIELD(table_name, 'department', 'instructor', 'student',
                'student_phone', 'course', 'enrollment'), constraint_type;

SELECT constraint_name, check_clause
FROM   information_schema.check_constraints
WHERE  constraint_schema = 'college_db'
ORDER BY constraint_name;

```

Q7: Insert valid rows (all constraints satisfied)

```

INSERT INTO department (dept_id, dept_code, dept_name, building, budget) VALUES
    (10, 'CSE', 'Computer Engineering', 'A Block', 2500000),
    (20, 'AIML', 'AI and Machine Learning', 'B Block', 3000000);
INSERT INTO department (dept_id, dept_code, dept_name)
VALUES (30, 'ECE', 'Electronics');
INSERT INTO instructor VALUES
    (101, 'Anil Kulkarni', 'Professor', 10, 185000, 'anil.k@example.com', NULL),
    (103, 'Rahul Patil', 'Asst. Prof', 10, 82000, 'rahul.p@example.com', 101);
INSERT INTO student (roll_no, student_name, gender, dob, dept_id, email) VALUES
    ('S24CS001', 'Aarav Sharma', 'M', '2005-03-14', 10, 'aarav@example.com'),
    ('S24AI001', 'Rohan Mehta', 'M', '2005-01-09', 20, 'rohan@example.com');
INSERT INTO student_phone VALUES
    (1, '9876500011'), (1, '9876500012'), (2, '9876500031');
INSERT INTO course VALUES ('CS101', 'Programming in C', 10, 4, 103, NULL),
    ('CS201', 'Data Structures', 10, 4, 103, 'CS101');
INSERT INTO enrollment VALUES (1, 'CS101', 3, 78, 'A'), (2, 'CS101', 3, 65, 'B');
SELECT * FROM department;
SELECT student_id, roll_no, student_name, dept_id, city FROM student;

```

Department 30 is inserted without building and budget, so building becomes NULL and budget takes its DEFAULT value. The students are inserted without student_id (generated by AUTO_INCREMENT) and without city (DEFAULT 'Pune').

Q8: Test every constraint with invalid data (each statement must be rejected)

```

INSERT INTO department VALUES (10, 'MECH', 'Mechanical', 'D Block', 1500000);
INSERT INTO department VALUES (40, 'CSE', 'Computer Science', 'A Block', 900000);
INSERT INTO department VALUES (40, 'MECH', NULL, 'D Block', 1500000);
INSERT INTO department VALUES (40, 'MECH', 'Mechanical', 'D Block', -500);
INSERT INTO instructor VALUES (104, 'Sneha Deshmukh', 'Assoc. Prof', 20, 15000,
    'sneha.d@example.com', NULL);
INSERT INTO instructor VALUES (104, 'Sneha Deshmukh', 'Assoc. Prof', 99, 115000,
    'sneha.d@example.com', NULL);
INSERT INTO student (roll_no, student_name, gender, dob, dept_id, email)
VALUES ('S24CS009', 'Test Student', 'X', '2005-01-01', 10, 'test@example.com');
INSERT INTO student (roll_no, student_name, gender, dob, dept_id, email)
VALUES ('S24CS009', 'Test Student', 'M', '1990-01-01', 10, 'test@example.com');
INSERT INTO student_phone VALUES (1, '12345');
INSERT INTO course VALUES ('CS301', 'DBMS', 10, 7, 101, 'CS201');
INSERT INTO enrollment VALUES (1, 'CS101', 3, 80, 'A');
INSERT INTO enrollment VALUES (1, 'CS201', 3, 120, NULL);
DELETE FROM department WHERE dept_id = 10;
DROP TABLE department;

```

Q9: Referential actions: ON UPDATE CASCADE and ON DELETE CASCADE / SET NULL

```
UPDATE department SET dept_id = 11 WHERE dept_id = 10;
SELECT roll_no, dept_id FROM student;
SELECT instr_id, instr_name, dept_id FROM instructor;
DELETE FROM student WHERE student_id = 1;
SELECT * FROM student_phone;
SELECT * FROM enrollment;
DELETE FROM instructor WHERE instr_id = 103;
SELECT course_id, title, instr_id FROM course;
```

Q10: Modify constraints with ALTER TABLE

```
ALTER TABLE course
  ADD CONSTRAINT chk_course_id CHECK (course_id REGEXP '^[A-Z]{2}[0-9]{3}$');
INSERT INTO course VALUES ('cs-99', 'Bad Course Id', 11, 3, NULL, NULL);
ALTER TABLE instructor
  ADD COLUMN joining_date DATE NOT NULL DEFAULT '2020-06-01';
ALTER TABLE instructor DROP CHECK chk_salary;
ALTER TABLE instructor
  ADD CONSTRAINT chk_salary CHECK (salary BETWEEN 40000 AND 300000);
ALTER TABLE student MODIFY city VARCHAR(30) NOT NULL DEFAULT 'Pune';
ALTER TABLE student DROP INDEX uq_student_email;
ALTER TABLE student ADD CONSTRAINT uq_student_email UNIQUE (email);
DESCRIBE instructor;
```

8. Sample Input / Database Tables

Constraint test data used in Q8 is given with the queries. The valid rows inserted in Q7 (the state of the tables before the violation tests) are shown below; the values were read from the database after executing Q1-Q7.

Table: department

dept_id	dept_code	dept_name	building	budget
10	CSE	Computer Engineering	A Block	2500000.00
20	AIML	AI and Machine Learning	B Block	3000000.00
30	ECE	Electronics	NULL	1000000.00

Table: instructor

instr_id	instr_name	designation	dept_id	salary	email	mentor_id
101	Anil Kulkarni	Professor	10	185000.00	anil.k@example.com	NULL
103	Rahul Patil	Asst. Prof	10	82000.00	rahul.p@example.com	101

Table: student

student_id	roll_no	student_name	gender	dob	dept_id	city	email
1	S24CS001	Aarav Sharma	M	2005-03-14	10	Pune	aarav@example.com
2	S24AI001	Rohan Mehta	M	2005-01-09	20	Pune	rohan@example.com

Table: student_phone

student_id	phone
1	9876500011
1	9876500012
2	9876500031

Table: course

course_id	title	dept_id	credits	instr_id	prereq_id
CS101	Programming in C	10	4	103	NULL
CS201	Data Structures	10	4	103	CS101

Table: enrollment

student_id	course_id	semester	marks	grade
1	CS101	3	78	A
2	CS101	3	65	B+

9. Output

The output below is the actual text produced by the MySQL 8.0.46 command-line client when the statements of Section 7 were executed in batch mode (`mysql --table -vvv < script.sql`). Each result is preceded by the statement that produced it; statements longer than a few lines are abbreviated with '...'. Because the script ran in batch mode, error messages report the line number within the script file. Execution times shown in brackets depend on the machine.

Output of Q1: Create the database

```
mysql> DROP DATABASE IF EXISTS college_db;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> CREATE DATABASE college_db;
Query OK, 1 row affected (0.00 sec)
```

Output of Q2: DEPARTMENT table: PRIMARY KEY, NOT NULL, UNIQUE, CHECK and DEFAULT

```
mysql> CREATE TABLE department (
->   dept_id      INT          NOT NULL,
->   dept_code   VARCHAR(10)  NOT NULL,
->   dept_name   VARCHAR(40)  NOT NULL,
->   ...
-> );
Query OK, 0 rows affected (0.01 sec)
```

Output of Q3: INSTRUCTOR table: FOREIGN KEY with referential actions and self-reference

```
mysql> CREATE TABLE instructor (
->   instr_id     INT          NOT NULL,
->   instr_name   VARCHAR(40)  NOT NULL,
->   designation  VARCHAR(20)  NOT NULL,
->   ...
-> );
Query OK, 0 rows affected (0.02 sec)
```

Output of Q4: STUDENT and STUDENT_PHONE tables: AUTO_INCREMENT, CHECK with IN and REGEXP

```
mysql> CREATE TABLE student (
->     student_id    INT                NOT NULL AUTO_INCREMENT,
->     roll_no       VARCHAR(10)        NOT NULL,
->     student_name   VARCHAR(40)       NOT NULL,
->     ...
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE student_phone (
->     student_id    INT                NOT NULL,
->     phone         CHAR(10)          NOT NULL,
->     CONSTRAINT pk_student_phone PRIMARY KEY (student_id, phone),
->     CONSTRAINT chk_phone CHECK (phone REGEXP '^[6-9][0-9]{9}$'),
->     CONSTRAINT fk_phone_student FOREIGN KEY (student_id)
->         REFERENCES student (student_id) ON DELETE CASCADE
-> );
Query OK, 0 rows affected (0.02 sec)
```

Output of Q5: COURSE and ENROLLMENT tables: composite PRIMARY KEY and ON DELETE CASCADE

```
mysql> CREATE TABLE course (
->     course_id     CHAR(5)           NOT NULL,
->     title          VARCHAR(40)      NOT NULL,
->     dept_id       INT               NOT NULL,
->     ...
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE enrollment (
->     student_id    INT               NOT NULL,
->     course_id     CHAR(5)          NOT NULL,
->     semester      TINYINT          NOT NULL,
->     ...
-> );
Query OK, 0 rows affected (0.02 sec)
```

Output of Q6: View all constraints from the data dictionary

```
mysql> SELECT table_name, constraint_name, constraint_type
-> FROM    information_schema.table_constraints
-> WHERE   table_schema = 'college_db'
-> ORDER BY FIELD(table_name, 'department', 'instructor', 'student',
->                 'student_phone', 'course', 'enrollment'), constraint_type;
```

TABLE_NAME	CONSTRAINT_NAME	CONSTRAINT_TYPE
department	chk_budget	CHECK
department	PRIMARY	PRIMARY KEY
department	uq_dept_code	UNIQUE
instructor	chk_salary	CHECK
instructor	fk_instr_mentor	FOREIGN KEY
instructor	fk_instr_dept	FOREIGN KEY
instructor	PRIMARY	PRIMARY KEY
instructor	uq_instr_email	UNIQUE
student	chk_gender	CHECK
student	chk_dob	CHECK
student	fk_student_dept	FOREIGN KEY
student	PRIMARY	PRIMARY KEY
student	uq_student_email	UNIQUE
student	uq_roll_no	UNIQUE
student_phone	chk_phone	CHECK
student_phone	fk_phone_student	FOREIGN KEY
student_phone	PRIMARY	PRIMARY KEY
course	chk_credits	CHECK
course	fk_course_prereq	FOREIGN KEY

course	fk_course_instr	FOREIGN KEY
course	fk_course_dept	FOREIGN KEY
course	PRIMARY	PRIMARY KEY
course	uq_course_title	UNIQUE
enrollment	chk_semester	CHECK
enrollment	chk_marks	CHECK
enrollment	fk_enr_student	FOREIGN KEY
enrollment	fk_enr_course	FOREIGN KEY
enrollment	PRIMARY	PRIMARY KEY

28 rows in set (0.00 sec)

```
mysql> SELECT constraint_name, check_clause
-> FROM information_schema.check_constraints
-> WHERE constraint_schema = 'college_db'
-> ORDER BY constraint_name;
```

CONSTRAINT_NAME	CHECK_CLAUSE
chk_budget	(`budget` > 0)
chk_credits	(`credits` between 1 and 5)
chk_dob	(`dob` >= _utf8mb4'1995-01-01')
chk_gender	(`gender` in (_utf8mb4'M',_utf8mb4'F',_utf8mb4'O'))
chk_marks	(`marks` between 0 and 100)
chk_phone	regexp_like(`phone`,_utf8mb4'^[6-9][0-9]{9}\$')
chk_salary	(`salary` between 30000 and 300000)
chk_semester	(`semester` between 1 and 8)

8 rows in set (0.00 sec)

Output of Q7: Insert valid rows (all constraints satisfied)

```
mysql> INSERT INTO department (dept_id, dept_code, dept_name, building, budget) VALUES
-> (10, 'CSE', 'Computer Engineering', 'A Block', 2500000),
-> (20, 'AIML', 'AI and Machine Learning', 'B Block', 3000000);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO department (dept_id, dept_code, dept_name)
-> VALUES (30, 'ECE', 'Electronics');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO instructor VALUES
-> (101, 'Anil Kulkarni', 'Professor', 10, 185000, 'anil.k@example.com', NULL),
-> (103, 'Rahul Patil', 'Asst. Prof', 10, 82000, 'rahul.p@example.com', 101);
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO student (roll_no, student_name, gender, dob, dept_id, email) VALUES
-> ('S24CS001', 'Aarav Sharma', 'M', '2005-03-14', 10, 'aarav@example.com'),
-> ('S24AI001', 'Rohan Mehta', 'M', '2005-01-09', 20, 'rohan@example.com');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO student_phone VALUES
-> (1, '9876500011'), (1, '9876500012'), (2, '9876500031');
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO course VALUES ('CS101', 'Programming in C', 10, 4, 103, NULL),
-> ('CS201', 'Data Structures', 10, 4, 103, 'CS101');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO enrollment VALUES (1, 'CS101', 3, 78, 'A'), (2, 'CS101', 3, 65, 'B+');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM department;
+-----+-----+-----+-----+-----+
| dept_id | dept_code | dept_name | building | budget |
+-----+-----+-----+-----+-----+
| 10 | CSE | Computer Engineering | A Block | 2500000 |
| 20 | AIML | AI and Machine Learning | B Block | 3000000 |
| 30 | ECE | Electronics |  |  |
+-----+-----+-----+-----+-----+
```

```
|      10 | CSE      | Computer Engineering | A Block | 2500000.00 |
|      20 | AIML     | AI and Machine Learning | B Block | 3000000.00 |
|      30 | ECE      | Electronics          | NULL   | 1000000.00 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT student_id, roll_no, student_name, dept_id, city FROM student;
+-----+-----+-----+-----+-----+
| student_id | roll_no | student_name | dept_id | city |
+-----+-----+-----+-----+-----+
|          1 | S24CS001 | Aarav Sharma |      10 | Pune |
|          2 | S24AI001 | Rohan Mehta  |      20 | Pune |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Output of Q8: Test every constraint with invalid data (each statement must be rejected)

```
mysql> INSERT INTO department VALUES (10, 'MECH', 'Mechanical', 'D Block', 1500000);
ERROR 1062 (23000) at line 122: Duplicate entry '10' for key 'department.PRIMARY'

mysql> INSERT INTO department VALUES (40, 'CSE', 'Computer Science', 'A Block', 9000000);
ERROR 1062 (23000) at line 123: Duplicate entry 'CSE' for key 'department.uq_dept_code'

mysql> INSERT INTO department VALUES (40, 'MECH', NULL, 'D Block', 1500000);
ERROR 1048 (23000) at line 124: Column 'dept_name' cannot be null

mysql> INSERT INTO department VALUES (40, 'MECH', 'Mechanical', 'D Block', -500);
ERROR 3819 (HY000) at line 125: Check constraint 'chk_budget' is violated.

mysql> INSERT INTO instructor VALUES (104, 'Sneha Deshmukh', 'Assoc. Prof', 20, 15000,
->                                     'sneha.d@example.com', NULL);
ERROR 3819 (HY000) at line 126: Check constraint 'chk_salary' is violated.

mysql> INSERT INTO instructor VALUES (104, 'Sneha Deshmukh', 'Assoc. Prof', 99, 115000,
->                                     'sneha.d@example.com', NULL);
ERROR 1452 (23000) at line 128: Cannot add or update a child row: a foreign key constraint
fails (`college_db`.`instructor`, CONSTRAINT `fk_instr_dept` FOREIGN KEY (`dept_id`)
REFERENCES `department` (`dept_id`) ON DELETE SET NULL ON UPDATE CASCADE)

mysql> INSERT INTO student (roll_no, student_name, gender, dob, dept_id, email)
-> VALUES ('S24CS009', 'Test Student', 'X', '2005-01-01', 10, 'test@example.com');
ERROR 3819 (HY000) at line 130: Check constraint 'chk_gender' is violated.

mysql> INSERT INTO student (roll_no, student_name, gender, dob, dept_id, email)
-> VALUES ('S24CS009', 'Test Student', 'M', '1990-01-01', 10, 'test@example.com');
ERROR 3819 (HY000) at line 132: Check constraint 'chk_dob' is violated.

mysql> INSERT INTO student_phone VALUES (1, '12345');
ERROR 3819 (HY000) at line 134: Check constraint 'chk_phone' is violated.

mysql> INSERT INTO course VALUES ('CS301', 'DBMS', 10, 7, 101, 'CS201');
ERROR 3819 (HY000) at line 135: Check constraint 'chk_credits' is violated.

mysql> INSERT INTO enrollment VALUES (1, 'CS101', 3, 80, 'A+');
ERROR 1062 (23000) at line 136: Duplicate entry '1-CS101' for key 'enrollment.PRIMARY'

mysql> INSERT INTO enrollment VALUES (1, 'CS201', 3, 120, NULL);
ERROR 3819 (HY000) at line 137: Check constraint 'chk_marks' is violated.

mysql> DELETE FROM department WHERE dept_id = 10;
ERROR 1451 (23000) at line 138: Cannot delete or update a parent row: a foreign key
constraint fails (`college_db`.`course`, CONSTRAINT `fk_course_dept` FOREIGN KEY
(`dept_id`) REFERENCES `department` (`dept_id`) ON UPDATE CASCADE)

mysql> DROP TABLE department;
ERROR 3730 (HY000) at line 139: Cannot drop table 'department' referenced by a foreign key
constraint 'fk_instr_dept' on table 'instructor'.
```

Output of Q9: Referential actions: ON UPDATE CASCADE and ON DELETE CASCADE / SET NULL

```
mysql> UPDATE department SET dept_id = 11 WHERE dept_id = 10;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT roll_no, dept_id FROM student;
+-----+-----+
| roll_no | dept_id |
+-----+-----+
| S24CS001 |      11 |
| S24AI001 |      20 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT instr_id, instr_name, dept_id FROM instructor;
+-----+-----+-----+
| instr_id | instr_name | dept_id |
+-----+-----+-----+
|      101 | Anil Kulkarni |      11 |
|      103 | Rahul Patil |      11 |
+-----+-----+-----+
2 rows in set (0.01 sec)
```

```
mysql> DELETE FROM student WHERE student_id = 1;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM student_phone;
+-----+-----+
| student_id | phone |
+-----+-----+
|          2 | 9876500031 |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM enrollment;
+-----+-----+-----+-----+-----+
| student_id | course_id | semester | marks | grade |
+-----+-----+-----+-----+-----+
|          2 | CS101 |          3 |      65 | B+ |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> DELETE FROM instructor WHERE instr_id = 103;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT course_id, title, instr_id FROM course;
+-----+-----+-----+
| course_id | title | instr_id |
+-----+-----+-----+
| CS101 | Programming in C | NULL |
| CS201 | Data Structures | NULL |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

Output of Q10: Modify constraints with ALTER TABLE

```
mysql> ALTER TABLE course
-> ADD CONSTRAINT chk_course_id CHECK (course_id REGEXP '^[A-Z]{2}[0-9]{3}$');
Query OK, 2 rows affected (0.05 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO course VALUES ('cs-99', 'Bad Course Id', 11, 3, NULL, NULL);
ERROR 3819 (HY000) at line 152: Check constraint 'chk_course_id' is violated.
```

```
mysql> ALTER TABLE instructor
-> ADD COLUMN joining_date DATE NOT NULL DEFAULT '2020-06-01';
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```

mysql> ALTER TABLE instructor DROP CHECK chk_salary;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE instructor
-> ADD CONSTRAINT chk_salary CHECK (salary BETWEEN 40000 AND 300000);
Query OK, 1 row affected (0.04 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE student MODIFY city VARCHAR(30) NOT NULL DEFAULT 'Pune';
Query OK, 1 row affected (0.05 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE student DROP INDEX uq_student_email;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE student ADD CONSTRAINT uq_student_email UNIQUE (email);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESCRIBE instructor;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| instr_id       | int           | NO   | PRI | NULL    |       |
| instr_name     | varchar(40)   | NO   |     | NULL    |       |
| designation    | varchar(20)   | NO   |     | NULL    |       |
| dept_id        | int           | YES  | MUL | NULL    |       |
| salary         | decimal(10,2) | NO   |     | NULL    |       |
| email          | varchar(40)   | NO   | UNI | NULL    |       |
| mentor_id      | int           | YES  | MUL | NULL    |       |
| joining_date   | date          | NO   |     | 2020-06-01 |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

```

10. Result

All six tables of *college_db* were created with 6 primary keys, 5 unique keys, 8 named CHECK constraints, 9 foreign keys, NOT NULL and DEFAULT constraints. Valid rows were accepted, while each of the 14 invalid statements in Q8 was rejected with the appropriate MySQL error (1062, 1048, 3819, 1452, 1451 and 3730). ON UPDATE CASCADE, ON DELETE CASCADE and ON DELETE SET NULL behaved as declared, and constraints were successfully added, dropped and modified using ALTER TABLE.

11. Conclusion

Declaring constraints in the schema moves data validation from application programs into the DBMS, so that every program and user is forced to obey the same rules. Primary keys provide entity integrity, foreign keys provide referential integrity and CHECK, UNIQUE, NOT NULL and DEFAULT provide domain integrity. Naming constraints makes violations easy to understand and allows them to be changed later with ALTER TABLE. CHECK constraints are enforced only from MySQL 8.0.16 onwards, which must be kept in mind when older servers are used.